Geometry

A Modern Introduction

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Contents

Preface

Geometry is the study of the properties and relationships of figures in the plane and in space. Since the time of Euclid, it has served as one of the principal means by which students learn the nature of logical reasoning and mathematical proof. From a small collection of definitions, postulates, and previously established results, an extensive body of knowledge is developed through careful deduction.

The purpose of this book is to present the fundamental principles of Euclidean geometry in a clear, orderly, and systematic manner. Each topic is introduced with precise definitions and illustrated by examples and diagrams before being reinforced through exercises of varying difficulty. Throughout the text, emphasis is placed not only on obtaining correct answers but also on understanding the reasoning that justifies them.

Geometry is more than the study of shapes and measurements. It provides a framework for analyzing patterns, making logical arguments, and solving problems with precision. These skills are of lasting value in mathematics, the sciences, engineering, architecture, and many other disciplines.

The reader is encouraged to approach each new topic thoughtfully, to examine every diagram carefully, and to develop the habit of proving statements rather than accepting them by intuition alone. Mastery of geometry is achieved through patient study, regular practice, and an appreciation for the logical structure that unifies the subject.

It is hoped that this book will provide a firm foundation in geometry and foster an enduring appreciation for the elegance, rigor, and usefulness of mathematical reasoning.

Chapter 1

Essentials of Geometry

Perimeter: The distance around a rectangle is called its **perimeter.**

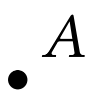
Circumference: The distance around a circle is called its **circumference.**

1.1 Points, Lines, and Planes

UNDEFINED TERMS

Undefined terms: In geometry, the words *point*, *line*, and *plane* are undefined terms. These words do not have formal definitions, but there is agreement about what they mean.

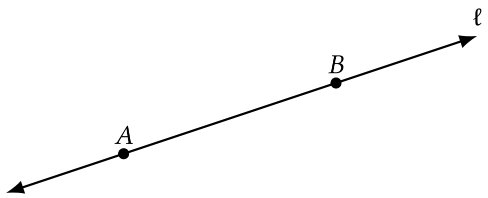
Point: A point has no dimension. It is represented by a dot.



point A

Line: A line has one dimension. It is represented by a line with two arrowheads, but it extends without end.

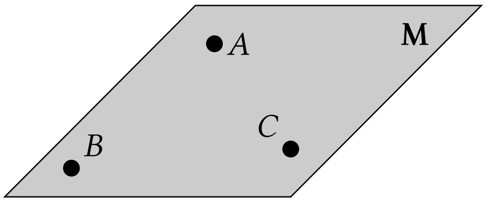
Through any two points, there is exactly one line. You can use any two points on a line to name it.



line ℓ, line *AB* ($\overleftrightarrow{AB}$), or line *BA* ($\overleftrightarrow{BA}$)

Plane: A plane has two dimensions. It is represented by a shape that looks like a floor or a wall, but it extends without end.

Through any three points not on the same line, there is exactly one plane. You can use three points that are not all on the same line to name a plane.



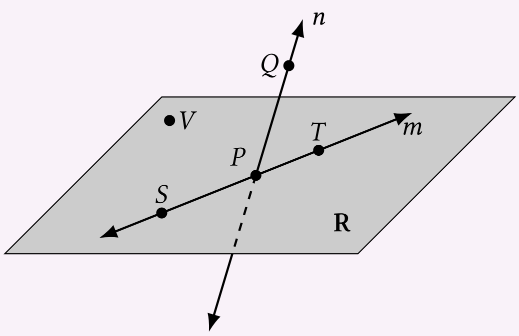
plane **M** or plane *ABC*

Collinear points: Collinear points are points that lie on the same line.

Coplanar points: Coplanar points are points that lie in the same plane.

Example 1

Name points, lines, and planes



(a) Give two other names for $\overleftrightarrow{PQ}$

(b) Give two other names for plane **R**.

(c) Name three points that are collinear.

(d) Name four points that are coplanar.

Solution

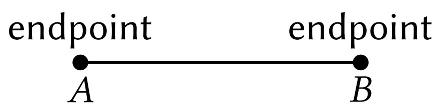
(a) Line n, $\overleftrightarrow{QP}$.
(b) Plane VST, plane VPT.
(c) Points S, P, and T.
(d) Points V, S, P, and T.

DEFINED TERMS

Defined terms: In geometry, terms that can be described using known words such as *point* or *line* are called defined terms.

Segment: The line segment AB, or segment AB, (written as $\overline{AB}$) consists of the endpoints A and B and all points on $\overleftrightarrow{AB}$ that are between A and B.

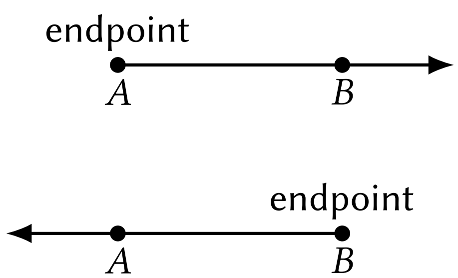
Note that $\overline{AB}$ can also be named $\overline{BA}$.



segment $\overline{AB}$ or $\overline{BA}$

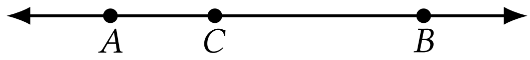
Ray: The ray AB (written as $\overrightarrow{AB}$) consists of the point A and all points on $\overleftrightarrow{AB}$ that lie on the same side of A as B.

Note that $\overrightarrow{AB}$ and $\overrightarrow{BA}$ are different rays.



rays $\overrightarrow{AB}$ and $\overrightarrow{BA}$

Opposite rays: If point C lies on $\overleftrightarrow{AB}$ between A and B, then $\overrightarrow{CA}$ and $\overrightarrow{CB}$ are opposite rays.

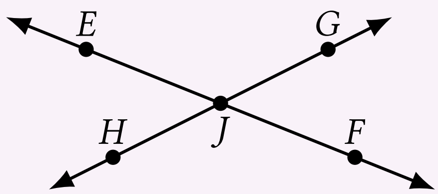


opposite rays $\overrightarrow{CA}$ and $\overrightarrow{CB}$

Segments and rays are collinear if they lie on the same line. So, opposite rays are collinear. Lines, segments, and rays are coplanar if they lie in the same plane.

Example 2

Name segments, rays, and opposite rays



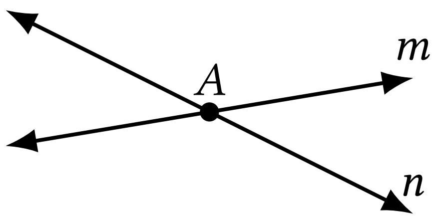
(a) Give another name for $\overline{GH}$.
(b) Name all rays with endpoint J. Which of these rays are opposite rays?

Solution

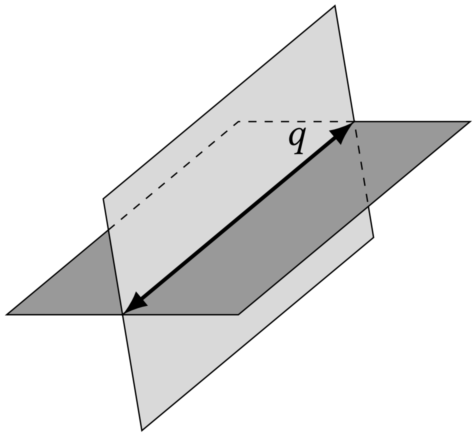
(a) Another name for $\overrightarrow{GH}$ is $\overrightarrow{HG}$.
(b) The rays with endpoint J are $\overrightarrow{JE}$, $\overrightarrow{JG}$, $\overrightarrow{JF}$, and $\overrightarrow{JH}$.
The pairs of opposite rays with endpoint J are $\overrightarrow{JE}$ and $\overrightarrow{JF}$, and $\overrightarrow{JG}$ and $\overrightarrow{JH}$.

INTERSECTIONS

Intersection: Two or more geometric figures intersect if they have one or more points in common. The intersection of the figures is the set of points the figures have in common. Some examples of intersections are shown below.



The intersection of two different lines is a point.



The intersection of two different planes is a line.

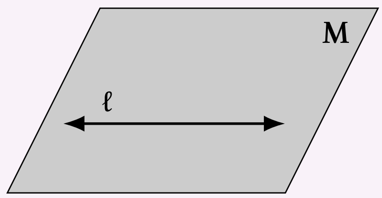
Example 3

Sketch intersections of lines and planes

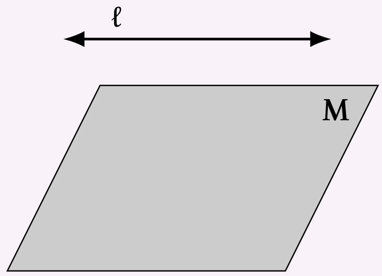
(a) Sketch a plane and a line that is in the plane.
(b) Sketch a plane and a line that does not intersect the plane.
(c) Sketch a plane and a line that intersects the plane at a point.

Solution

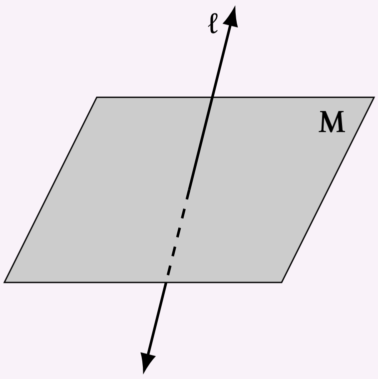
(a) The line ℓ is in the plane **M**.



(b) The line ℓ does not intersect the plane **M**.



(c) The line ℓ intersects the plane **M** at a point.



Outside box: 7.25pt plus 1.45009pt minus 1.45009pt

Example 4

Sketch intersections of planes

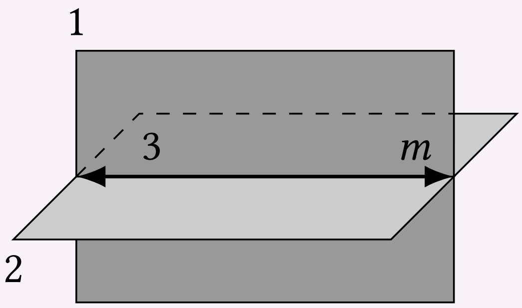
Sketch two planes that intersect in a line.

Solution

step 1 Draw a vertical plane. Shade the plane.

step 2 Draw a second plane that is horizontal. Shade this plane a different color. Use dashed lines to show where one plane is hidden.

step 3 Draw the line of intersection.



1.2 Use Segments and Congruence

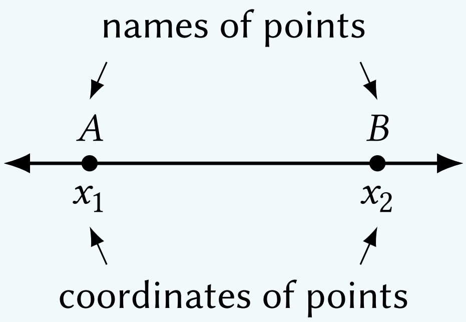
Postulate: In Geometry, a rule that is accepted without proof is called a postulate or axiom.

Theorem: A rule that can be proved is called a theorem.

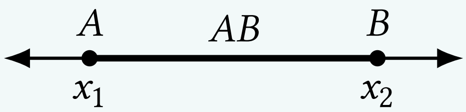
Postulate 1

Ruler Postulate

The points on a line can be matched one to one with the real numbers. The real number that corresponds to a point is the **coordinate** of the point.



The **distance** between points *A* and *B*, written as *AB*, is the absolute value of the difference of the coordinates of *A* and *B*.

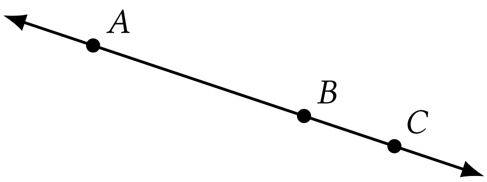


$$AB = |x_2 - x_1|$$

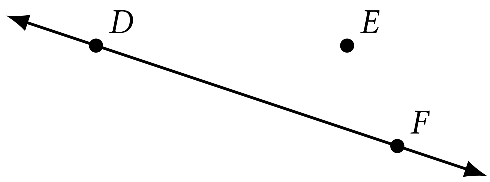
The distance between points *A* and *B*, or *AB*, is also called the *length* of $\overline{AB}$.

ADDING SEGMENT LENGTHS

When three points are collinear, you can say that one point is **between** the other two.



Point *B* is between points *A* and *C*.

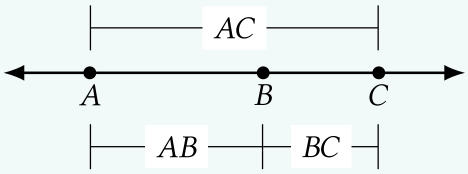


Point *E* is not between points *D* and *F*.

Postulate 2

Segment Addition Postulate

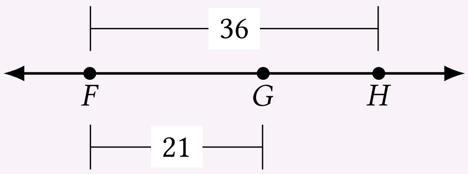
If B is between A and C, then $AB + BC = AC$. If $AB + BC = AC$, then B is between A and C.



Example 1

Find a length

Use the diagram to find GH.

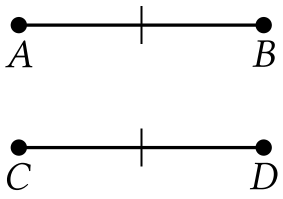


SolutionUse the Segment Addition Postulate to write an equation. Then solve the equation to find GH.

$FH = FG + GH$	Segment Addition Postulate
$36 = 21 + GH$	Substitute 36 for FH and 21 for FG.
$15 = GH$	Subtract 21 from each side.

CONGRUENT SEGMENTS

Line segments that have the same length are called **congruent segments**. In the diagram below, you can say "the length of $\overline{AB}$ is equal to the length of $\overline{CD}$," or you can say "$\overline{AB}$ is congruent to $\overline{CD}$." The symbol $\cong$ means "is congruent to."



Lengths are equal.
$AB = CD$
↑
”is equal to”

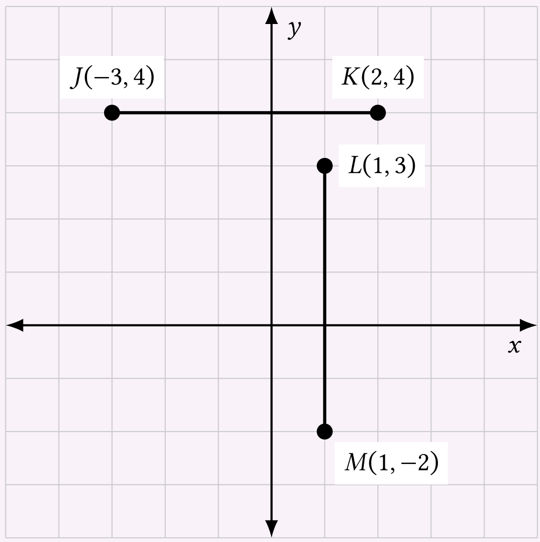
Segments are congruent.
$\overline{AB} \cong \overline{CD}$
↑
”is congruent to”

Example 2

Compare segments for congruence

Plot $J(23, 4)$, $K(2, 4)$, $L(1, 3)$, and $M(1, 22)$ in a coordinate plane. Then determine whether $\overline{JK}$ and $\overline{LM}$ are congruent.

Solution



To find the length of a horizontal segment, find the absolute value of the difference of the x-coordinates of the endpoints.

$JK = |2 - (-3)| = 5$ Use Ruler Postulate.

To find the length of a vertical segment, find the absolute value of the difference of the y-coordinates of the endpoints.

$LM = | - 2 - 3| = 5$ Use Ruler Postulate.

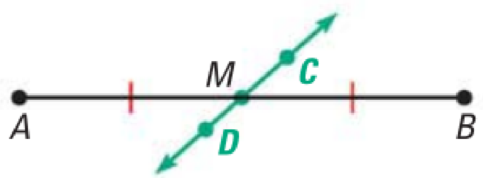
JK and LM have the same length. So, $\overline{JK} \cong \overline{LM}$.

1.3 Use Midpoint and Distance Formulas

MIDPOINTS AND BISECTORS

The **midpoint** of a segment is the point that divides the segment into two congruent segments. A **segment bisector** is a point, ray, line, line segment, or plane th at intersects the segment at its midpoint. A midpoint or a segment bisector *bisects* a segment.

M is the midpoint of $\overline{AB}$. So $\overline{AM} \cong \overline{MB}$ and $AM = MB$.

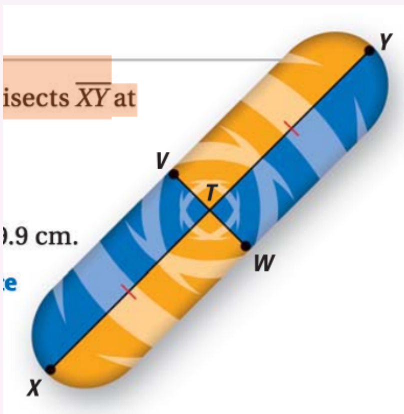


CD is a segment bisector of AB. So, $\overline{AM} \cong \overline{MB}$ and $AM = MB$.

Example 1

Find segment lengths

In the skateboard design, VW bisects XY at point T, and $XT = 39.9$ cm. Find XY.



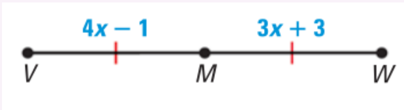
SolutionPoint T is the midpoint of XY. So, $XT = TY = 39.9\,\text{cm}$.

$$\begin{aligned} XY &= XT + TY && \text{Segment Addition Postulate} \\ &= 39.9 + 39.9 \\ &= 79.8\text{cm} \end{aligned}$$

Example 2

Use algebra with segment lengths

Point M is the midpoint of $\overline{VW}$. Find the length of $\overline{VM}$.



Solution

step 1 Write and solve an equation. Use the fact that that $VM = MW$.

$$\begin{aligned} VM &= MW && \text{Write equation.} \\ 4x - 1 &= 3x - 3 && \text{Substitute.} \\ x - 1 &= 3 && \text{Subtract } 3x \text{ from each side.} \\ x &= 4 && \text{Add 1 to each side.} \end{aligned}$$

step 2 Evaluate the expression for VM when $x = 4$.

$$VM = 4x - 1 = 4(4) - 1 = 15$$

So the length of $\overline{VM}$ is $\boxed{15}$.

check Because $VM = MW$, the length of $\overline{MW}$ should be 15. If you evaluate the expression for MW, you should find that $MW = 15$.

$$MW = 3x + 3 = 3(4) + 3 = 15\checkmark$$

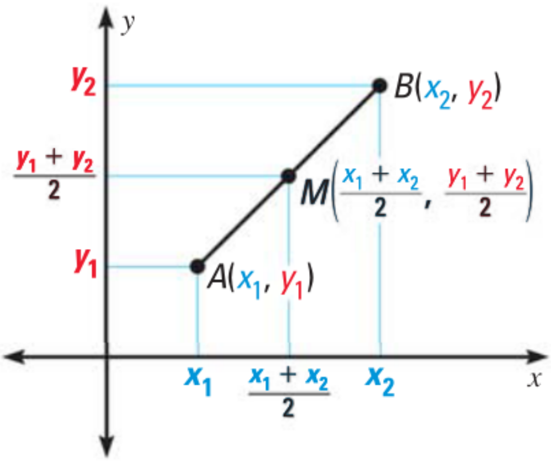
You can use the coordinates of the endpoints of a segment to find the coordinates of the midpoint.

THE MIDPOINT FORMULA

The coordinates of the midpoint of a segment are the averages of the x-coordinates and of the y-coordinates of the endpoints.

If $A(x_1, y_1)$ and $B(x_2, y_2)$ are points in a coordinate plane, then the midpoint M $\overline{AB}$ has coordinates

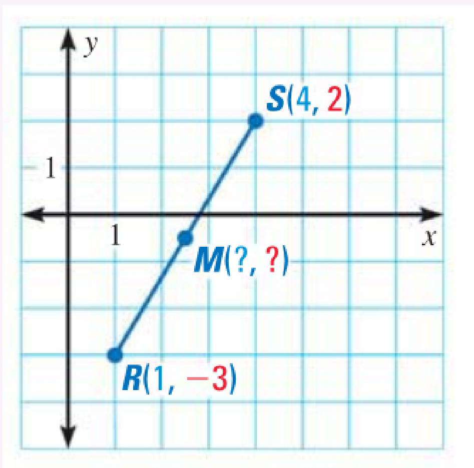
$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$



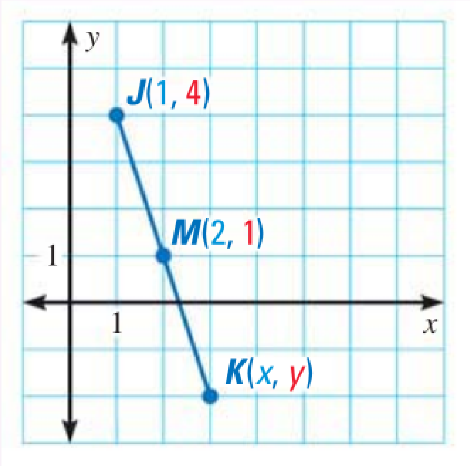
Example 3

Use the Midpoint Formula

(a) The endpoints of $\overline{RS}$ are $R(1, -3)$ and $S(4, 2)$. Find the coordinates of the midpoint M.



(b) The midpoint of $\overline{JK}$ is $M(2, 1)$. One endpoint is $J(1, 4)$. Find the coordinates of endpoint K.



Solution

(a) Use the Midpoint Formula.

$$M\left(\frac{1+4}{2}, \frac{-3+2}{2}\right) = M\left(\frac{5}{2}, -\frac{1}{2}\right)$$

The coordinates of the midpoint M are $\boxed{\left(\frac{5}{2}, -\frac{1}{2}\right)}$.

(b) Let (x, y) be the coordinates of endpoint K. Use the Midpoint Formula.
step 1 Find x.

$$\frac{1+x}{2} = 2$$
$$1 + x = 4$$
$$x = 3$$

step 2 Find y.

$$\frac{4+y}{2} = 1$$
$$4 + y = 2$$
$$y = -2$$

The coordinates of endpoint K are $\boxed{(3, -2)}$.

DISTANCE FORMULA

The Distance Formula is a formula for computing the distance between two points in a coordinate plane.

Appendix

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